Parametric Equations:

$$X = \frac{D}{2}\cos(\varphi) - \frac{dr}{2}\cos(\varphi + \gamma) - e\cos(N \cdot \varphi)$$

$$y = -\frac{D}{2}\sin(\varphi) + \frac{dr}{2}\sin(\varphi + \gamma) + e\sin(N \cdot \varphi)$$

$$\gamma = \tan^{-1}\left(\frac{\sin((N-1)\cdot\varphi)}{\cos((N-1)\cdot\varphi) - \frac{D}{2 \cdot e \cdot N}}\right)$$

D: pitch circle Diameter of roller plate.

dr: roller Diameter.

N: roller count.

e: eccentricity.

φ: parameter (angle) = $0 \ldots 2\pi \ldots$

contracted cycloid: the cutting milling center is placed at a distance: e ~~away~~ in ward from the roller center.